AI-BASED SMART STUDY PLANNER

Project Report

BCA Semester Project

Prepared from the supplied Java source files and Oracle database schema.

1. Introduction

AI-Based Smart Study Planner is a Java desktop application intended to help students organize subjects, exams, study tasks, study sessions, learning notes, and study progress. It uses a Swing-based user interface and stores structured information in Oracle Database through JDBC.

2. Objectives

• Provide student registration and login.

• Organize subjects and exam information.

• Create study plans and record study sessions.

• Track completed study hours and progress information.

• Store notes and study-material metadata.

• Support a practice-quiz flow using AI-generated questions and save quiz results.

3. Technology Stack

Technology	Purpose
Java / Swing	Desktop application and user interface
Oracle Database 21c XE	Relational data storage
JDBC / Oracle JDBC driver	Java-to-Oracle connectivity
Apache Tika	Text extraction from supported materials
Gemini API integration	AI question generation; requires valid configuration

4. System Modules

Registration and Login: Student account registration and login screens.

Subject Management: Subjects, difficulty, preparation percentage, and target percentage.

Exam Management: Exam name, date, time, and exam type.

Study Planning: Study dates, hours, tasks, and status.

Study Sessions: Session duration, optional topic/notes, and synchronization of duration into completed hours.

Progress Tracking: Completed hours and progress percentage by student and subject.

Notes: Titled notes and note content, optionally linked to topic/source material.

Study Materials: Material metadata and file path; supported files can be processed for text extraction.

AI Practice Quiz: Multiple-choice parsing, answer checking, score display, and saved quiz results.

Quiz History / Dashboard: Corresponding UI source files are present in the repository.

5. Database Design

The supplied Oracle schema contains 13 tables: AI_RECOMMENDATIONS, EXAMS, MCQS, NOTES, NOTIFICATIONS, PROGRESS, QUIZ_RESULTS, STUDENTS, STUDY_MATERIALS, STUDY_PLANS, STUDY_SESSIONS, SUBJECTS, and TOPICS. Primary keys and foreign keys connect students, subjects, topics, materials, plans, sessions, progress, quizzes, and recommendations.

The complete schema script is supplied as Database/schema.sql. It is based on the DBMS_METADATA output shared for the STUDYPLANNER schema; Oracle physical storage/index clauses were omitted for portability.

6. Application Architecture

The source is organized into UI forms (ui), database access classes (dao), domain classes (model), services (service), database configuration (config), and Gemini integration (ai). Main.java launches LoginForm on the Swing event-dispatch thread.

7. Configuration and Execution

• Install a compatible JDK and Oracle Database; the code expects the XEPDB1 service.

• Create/select the STUDYPLANNER schema and review Database/schema.sql before running it. Do not execute it over an existing populated schema.

• Configure Oracle JDBC and Apache Tika libraries in the IDE/build classpath.

• Set STUDYPLANNER_DB_PASSWORD as an environment variable.

• Set GEMINI_API_KEY for API-backed AI features; model access must be available.

• Run com.smartstudy.app.Main after dependencies and environment variables are configured.

8. Security Considerations

Database credentials and API keys should be supplied through environment variables and must not be committed to version control. Keep student records and database exports private.

9. Limitations and Future Enhancements

• Add reproducible build/dependency management (Maven or Gradle).

• Provide a setup script and new-machine database guide.

• Verify AI API model/configuration against the active provider account.

• Add automated tests and expected input/output documentation.

• Review password storage and use appropriate password hashing before real-world deployment.

10. Conclusion

The project combines student records, study planning, subject/exam management, session tracking, notes, study materials, progress information, and a quiz workflow in a Java Swing application backed by Oracle Database. Setup requires the matching schema, JDBC/Tika dependencies, and environment configuration.

Appendix: The uploaded ZIP's src/resources/schema.sql was empty. A populated schema is included in this submission package.